Logic, Language and Philosophy

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A Short Introduction to Standard Logic

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1
Introduction

1.1 Reasoning and logic

Logic, thus is the common understanding, is 'the science of reasoning'. But what exactly is reasoning? And in what sense is logic its science?

The term 'reasoning' covers a wide range of phenomena, from formal proofs in mathematics to the loose kind of connections we make in everyday conversations between observations, judgements, assumptions, and the like. A strictly ordered sequence of steps that derives, say, the Pythagorean theorem from the Euclidean axioms according to fixed rules counts as reasoning. But so does looking out the window, observing heavy rain clouds and deciding to take an umbrella when going out. Reasoning is done in language, but also 'in the mind', or with pictures, or in any other medium that allows us to represent information.

Reasoning often takes the form of argumentation where the goal is to convince an audience or readership to accept a particular conclusion by providing arguments for it. Argumentation is the core of science, of course, where we argue for a particular theory or hypothesis on the basis of experimental data or conceptual considerations. But argumentation also plays a prominent role outside science: in law and politics, in theology and speculative philosophy, in business and in the arts, in everyday conversations, —argumentation is everywhere.

Not all reasoning and argumentation is of the same nature. There is what is a called 'deductive' reasoning, where given the truth of some sentences, called the 'premisses', the truth of another, called the 'conclusion', necessarily follows. Mathematical reasoning is of this kind. But we also have what is called 'inductive' reasoning, where from a finite number of observations that function as premisses, we draw a general conclusion. This is often what we find in experimental and

observational science. Another differentiator is whether we consider the truth of premisses and conclusion, or their probability.

Given this variety of forms and functions of reasoning and argumentation, one might wonder in what sense logic is 'the science' of all that?

There is an everyday use of the term 'logic', more common perhaps now than in the past, where it serves to indicate a pattern of more or less necessary connections. Thus, one may find such expressions as 'the logic behind the measures taken by the government', or 'the logic behind his madness', or 'the strange logic of supply and demand in on-line trading'. These terms refer to connections between certain actions, rules, statements that follow a certain pattern. They serve to explain why things are how they are, to predict future events, and so on. This use of the term refers directly to the patterns themselves: it is the patterns that constitute the logic.

But the term 'logic' is also used in another sense, not to refer such patterns, but to the discipline that studies them. Thus we speak of 'Aristotelian logic' to refer to the works of the Greek philosopher Aristotle[1], of 'Mohist logic' to refer to works by the Chinese thinker Mozi and his school[2], of the 'mathematical logic' of Frege and Russell, and so on.[3] All these uses of 'logic' refer to systematic studies of certain reasoning patterns.

These two uses of the term are related: particular logical systems focus on particular patterns of reasoning. That is why they are often used more or less interchangeably. Thus 'Aristotelian logic' can refer to a particular set of reasoning patterns, as well as to Aristotle's systematic treatment of them.

1.2 Arguments and argument schemes

Not all aspects of argumentation fall within the scope of logic. Traditionally, a distinction is made between logic and rhetoric. Whereas logic focusses on the validity of arguments, rhetoric is concerned with their persuasive force. One could say that logic deals with arguments that are used to convince, and rhetoric with arguments that are meant to persuade. Thus rhetoric focusses more on psychological processes induced by arguments, whereas logic studies their systematic properties.

The central concept of any logic is that of *validity*. We distinguish between arguments that are valid and those that are not. Consider the argument in (1):

(1) a. Some employees are from abroad
b. All people from abroad need working permits
c. So, some employees need working permits

Here we have two premisses, (1a) and (1b), and a conclusion, (1c). Intuitively this a valid argument: in any situation in which (1a) and (1b) are both true, (1c) must be true as well.

Contrast this with (2):

(2) a. All employees are from abroad
b. All company owners are from abroad
c. So, some employees are company owners

This is not a valid argument, since we can imagine a situation in which (2a) and (2b) are both true, but (2c) is false. Note that also if all three, i.e., (2a), (2b) ànd (2c), are true, the argument as such is still flawed.

Now what distinguishes the valid argument (1) from the invalid (2)?

One way to find out is to change elements in the sentences making up these arguments and see whether that changes their status. For example, suppose we change 'employees' to 'bus drivers'. The argument in (1) then becomes that in (1′):

(1′) a. Some bus drivers are from abroad
b. All people from abroad need working permits
c. So, some bus drivers need working permits

That is still a valid argument, so apparently that the first premisse and the conclusion are about employees is not relevant. With similar substitutions, we can figure out that 'be from abroad' and 'need a working permit' are also not relevant: we can substitute these expressions by any other, and as long as we do so uniformly, we get another valid argument.

What we can *not* change, however, are the expressions 'some' and 'all'. For example, if we replace 'all' in (1b) by 'some' we get:

(3) a. Some employees are from abroad
b. Some people from abroad need working permits
c. So, some employees need working permits

This is no longer a valid argument.

By changing elements we can thus identify those on which the validity of an argument depends. These are called *logical constants*, and non-essential elements *logical variables*. The distinction is captured in the notion of an *argument scheme*. An argument scheme is a skeleton that is obtained from a concrete argument, by replacing anything that does not contribute to the validity by variables. Here we use A, B, C as variables that range over nouns and verbs. The argument scheme underlying (1), (1′) and other variants is (4):

(4) a. Some A are B
b. All B are C
c. Some A are C

Any uniform substitution of concrete expressions for the variables A, B, and C will result in a concrete valid argument.

What holds for valid arguments and valid argument schemes, also applies to invalid ones. By the same method, we can then conclude that the invalid argument (2) is an instantiation of the invalid argument scheme (5):

(5) a. All A are B
b. All C are B
c. Some A are C

The extraction of argument schemes from concrete arguments is a process of *formalisation*. Argument schemes such as (4) and (5) are formulated using a mixture of natural language and formal language expressions. As we shall see in modern logic natural language is done away with and completely formal languages are used to formulate argument schemes and study their logical properties.

1.3 The anatomy of a logical system

A logical system, or 'logic' for short, is characterised by three key ingredients.

First of all, there is the formal language it employs. The main elements here are specifications of the set of logical constants and that of the logical variables. The set of logical constants defines what the logic is about, its descriptive scope, the range of argument schemes that it is intended to cover. The specification of the logical constants

and logical variables is often called the *vocabulary* of the formal language. Then there is a set of *rules* that specify how starting from the vocabulary one can build the set of all the expressions of the formal language. This first ingredient is called the *syntax* of the logic.

But expressions as such are not enough, they need to be interpreted. This is the second ingredient of the logic. The logical constants are assigned fixed meanings, ones that account for the contributions they make to the validity of argument schemes. The meanings of logical variables are characterised in terms of the type of meaning they have. This gives us the meanings of the expressions that make up the vocabulary. But we also need to specify the meanings of all the complex expressions that together make up the formal language. To this end another set of rules is formulated. Corresponding to each syntactic rules that defines a way to combine expressions into more complex ones, there is a corresponding semantic rule that defines the meaning of the complex expressions in terms of those of its component parts. This specification is called the *semantics* of the logic.

The third and last ingredient of the definition of a logic is a specification of the intuitive concept of *validity*. As we shall see, there are a number of ways to define validity of argument schemes. The main opposition here is that between a semantic approach and a syntactic one. As the name suggests, a semantic definition of validity proceeds in terms of meaning. Such a definition corresponds to the intuitive concept that we used before: an argument is valid if whenever the premisses are true, the conclusion is also true. But one can also approach validity in terms of the concept of a proof: an argument is valid if we can derive the conclusion from the premissen in a formal proof.

A final word about the way in which syntax and semantics of the formal language of a logical system are related. Given a vocabulary the syntactic rules define what the expressions of the language are by specifying a finite number of ways of combining given expressions into new, more complex ones. The semantics follows this pattern exactly: starting from the meanings assigned to the expressions that make up the vocabulary, there is a semantic counterpart for each syntactic rule that specifies the meaning of the complex expression in terms of the meanings of its component parts. This syntax – semantics parallelism is characteristic for the large majority of formal languages employed not just in logic, but also in mathematics, computer science, and other

fields. This core conception of how syntax and semantics relate is often called the *principle of compositionality*: 'the meaning of a complex expressions is a function of the meaning of its parts and the way in which they are syntactically related'.

1.4 Logic as a many-splendored thing

'Logic' is in a sense a misleading term because, as we have seen, it is highly ambiguous. It refers to an academic discipline, i.e., to what one studies or takes courses in. But it can also be used to refer to general, normative principles that characterise a certain institution or form of behaviour, to particular ways of thinking, reasoning and argumentation. And 'logic' is the term that refers to formal systems that can be used to study all this.

The plural of 'formal systems' has to be acknowledged. Different formal systems study different types of reasoning and argumentation. Thus there are systems that focus on reasoning with temporal notions or modal ones, or with quantificational aspects. That is one source of multiplicity. But there is another one, which is that there is no one, 'correct' or 'true' formal system that characterises *the* one, correct form of reasoning. So even with the same coverage in terms of what aspects of reasoning are being investigated, there is often a range of alternatives that give different accounts of what the logic of that particular aspect is supposed to be. Thus we have different logics for propositional reasoning, for quantificational expressions, for modal and temporal notions, and so on.

This introduction focusses on a couple of core formal systems. If we can use a gastronomical metaphor, in order to appreciate the variety that a certain cuisine has to offer, it is best to start with learning how to prepare and appreciate a number of standard dishes. But a good cookbook will also provide some pointers to variations on those standards, and thus provide an entrance to the endless variety and depths that characterise a good cuisine.

Logic is such a cuisine: it contains a dazzling variety of formal systems, concepts, techniques, all with their associated meta-logical results. Some of these systems are there just for their own sake, like many parts of pure mathematics. Others are closely tied to very specific and particular applications. These applications as such can come from

a wide variety of fields: philosophy, linguistics, economics, psychology and cognitive science, mathematics, computer science, ...

This book will provide a first introduction to two key systems that play a role in any variety or application: classical propositional logic and classical predicate logic. But it will also contain pointers to variations and alternatives, to logics that deal with a wider range of concepts. As for applications, the focus will be on language and philosophy. These two are intimately related, for obvious reasons: philosophical analysis is concerned with fundamental concepts and these concepts are expressed in language. Of course, although, or perhaps because the connection between philosophy and language is intimate, it is not therefore obvious and unproblematic. Philosophical concerns require careful linguistic analysis, and in its turn, the analysis of language raises fundamental philosophical questions.

The details are intricate, numerous and fascinating. What follows will not be able to do justice to that, but it does aspire to give the reader a first taste of the role that logic can play here. That will be done in two ways. First of all, we will provide some examples of the application of logic to philosophical problems. Secondly, we will give pointers to philosophical problems that are raised by logic itself. In that way we hope to give the reader a first glimpse of the wide variety of tastes, smells and textures, that await them once they have trained their palate to appreciate the standard dishes of the logic cuisine.

2

Excursus: logic in philosophy

2.1 The topic

In the western tradition logic has always been an integral part of philosophy. Aristotle established logic as a systematic philosophical discipline along with and connected with other philosophical disciplines, such as metaphysics, epistemology, ethics. The basic structure remained in place for many centuries, although the contents and also the relative importance of the various elements varied. Logic was central in the scholasticism of the European middle ages, but lost some of its prominence with the advent of modern philosophy, in which epistemological questions came to occupy center stage.

By the eighteenth century logic was still a recognised part of the philosophical curriculum, but its importance for other philosophical disciplines was declining. The term 'logic' came to have a much wider use, but as the systematic study of formal aspects of reasoning, logic had become of lesser importance.

That changed with the advent of modern formal logic in the late nineteenth, early twentieth century. The limitations of the classical Aristotelian logic had long been acknowledged, but never adequately overcome. By making a radical break with natural language as the primary vehicle for philosophical analysis, modern logic provided a much more powerful set of systematic, formal tools that could be applied in a wide range of areas, including philosophy.

Thus a new age of intensive interaction between logic and other philosophical disciplines emerged. Modern analytic philosophy was shaped by the application of modern logic, and continues to thrive in virtue of the application of ever more sophisticated logical tools. The foundations of this tradition were laid by a number of authors, such as

De Morgan, Boole, Peirce, and, if not the most important definitely the most influential: Frege and Russell.

2.2 The authors

Gottlob Frege

Gottlob Frege (1848 – 1925) was in the first place a mathematician, who found himself forced to become a logician, and who counts as a philosopher of language only retrospectively. For Frege himself this variety of professional identities was not quite relevant: he always approached the problems that confronted him during his intellectual development from the same broad, philosophical perspective. That nowadays Frege is considered to be a logician and a philosopher of language is largely due to the fact that we now view these as separate disciplines, to which Frege happens to have made decisive contributions.

Frege the logician is the author of *Begriffsschrift*, published in 1879, which counts as the first systematic exposition of modern logic. But Frege turned logician because he had to: he was driven to his logical investigations by a mathematical motive. The background of that is the foundational crisis in mathematics. Where mathematics had served as the paradigm of certain knowledge for centuries,[1] the development of non-Euclidean geometries and the paradoxes in Cantor's set theory had instigated doubts about the very foundations of mathematical knowledge. Frege, who was particularly interested in the arithmetical foundations of analysis, tried to base arithmetic on logic: the so-called 'logicistic programme'. In order to carry out this programme a logic was needed that was adequate for reasoning about relations, i.e., a logic that could deal adequately with relational judgments with more than one quantified arguments.[2] Within the broadly Aristotelian logical framework, still prevalent at the time, the basic form of judgments was that of subject – predicate form, and that was inadequate to deal with relational judgments. The system that Frege presented in *Begriffsschrift* is the first fully developed system that is.[3]

The reason that Frege constructed a formal system of logic was that he felt that a proper foundation of mathematics could only be achieved on the basis of a logic in which mathematical reasoning could

be formalised in a completely explicit way. That he considered arithmetic as the most fundamental part of mathematics is, of course, not by accident: especially after the development of analytic geometry by Descartes, number theory was, and perhaps still is, considered the most fundamental mathematical discipline. Frege had an additional motive: in his resistance against the prevailing naturalistic empiricism of his time, he argued that every form of scientific empirical knowledge has to be based on induction and inductive reasoning, and that a proper understanding of induction can only be based on arithmetic. By showing that arithmetic is *a priori*, Frege intended to rebut the claim of radical empiricism that *all* knowledge is empirical, thus restoring Kant's epistemology.

At some point in his logical investigations Frege looked at more general issues regarding meaning, which brought him to look at language more generally, going beyond the formal languages of logic and mathematics. In this context he developed his broadly platonistic perspective, and it is this body of work that has earned him the title of 'Founding Father of Analytic Philosophy'. On this view, which has been developed and defended by authors such as Dummett, Geach, Austin and Anscombe, Frege is a direct ancestor of early Wittgenstein, and, together with Russell and G.E. Moore, one of the creators of what is called 'the Linguistic Turn'.[4] From this point of view the importance of Frege's work is mainly located in his semantic investigations, and the corresponding ontological views.

There is no denying that the influence of Frege on Russell and early Wittgenstein, and through them on the entire tradition of analytical philosophy, is real and profound. Nevertheless it is primarily a systematic appraisal, one that can be made only in hindsight: we can now see that Frege in fact played that role. But if we want to assess Frege in his historical context, then looking at him and his work in hindsight might very well distort the picture. That we nowadays value Frege's work mainly from an analytic perspective, or, even more narrowly, from the perspective of the philosophy of language, does not mean that he himself saw what he did in that way.[5] In the literature on Frege there are also authors who try to interpret him in his contemporary context, and who come to a different appraisal of the nature of the intellectual setting in which Frege himself situated his work. On this view, defended among others by Hans Sluga[6] the main

motivation of Frege's work is epistemological, rather than language-oriented. The dominant paradigm in German philosophy in the mid-nineteenth century was naturalism, not some form of Kantian idealism. This naturalism came with a radical form of empiricism,[7] which, according to Sluga, is the background against which we should interpret Frege's philosophy. Frege's aim was to re-instate Kant's form of anti-empiricism. Empirical laws, Frege argued, depend on induction. And induction presupposes arithmetic. If it can be shown that arithmetic is a priori, then this would vindicate Kant's claim that not all knowledge is empirical. Frege does differ from Kant in considering arithmetic to be analytic, whereas Kant was of the opinion that it is synthetic.[8] The reduction of arithmetic to logic plays a key role.

As was mentioned such a reduction presupposes a logic with adequate expressive power, and that is what Frege developed in his *Begriffsschrift* from 1879 (Frege, 1879). After completing that work he set himself the task of providing a satisfactory philosophical analysis of the concept of number, which he laid down in his *Grundlagen der Arithmetik*, which appeared in 1884 (Frege, 1884). A formal analysis of arithmetic in logical terms is the topic of the *Grundgesetze der Arithmetik*, of which the first volume appeared in 1895 (Frege, 1895). Then something unexpected happened: Russell discovered that the axioms that Frege used in the first volume allowed the derivation of a paradox. Frege made an attempt to repair this. His revision appeared in an appendix to the second volume of *Grundgesetze*, that was already in print when Frege was confronted with Russell's discovery. However, it was soon clear that his revision failed to solve the problem, and Frege abandoned the logicistic programme.[9]

Bertrand Russell

The influence of Bertrand Russell (1872 – 1970) on the development of western analytic philosophy can hardly be overestimated. Together with G.E. Moore he shaped philosophy in the UK and beyond by his influence on a number of famous students, including Ludwig Wittgenstein, and through his many writings on a wide range of topics. Some of his works addressed highly technical issues in the philosophy of mathematics, formal logic, epistemology and ontology, and were intended for a readership of specialists. But he also wrote for a more

general audience on topics in philosophy, politics, educational reform, marriage, and so on.

Russell was born in an aristocratic family, and received the education that was common at the time in those social circles. He studied mathematics in Cambridge, but also developed broad interests in philosophy, politics and social issues. (His first publication was on German social democracy.)

His early years were devoted to what has become know as the 'Revolt against Idealism', and to philosophy of mathematics and logic. Philosophy in England in the second half of the 19th century was heavily under the influence of British Idealism, a form of idealism that was promoted by, among others, Bosanquet and Bradley, and that had strong roots in the German idealism of Kant and Hegel. Russell initially was trained in this tradition of speculative metaphysics, but in the late nineties of the 19th century, together with G.E. Moore, rebelled against it. Initially, his main reason for rejecting idealism was its insistence that all relations are 'internal', i.e., can be understood in terms of properties of the things related. Through his training in mathematics, Russell saw that this could not work for asymmetrical relations in mathematics. For example, the relation of being smaller than between natural numbers can not be explained in terms of properties of the numbers themselves.

For Russell this led to a rejection of the basic tenets of idealism, including its insistence on a form of holism, and its view that thought and fact can not be distinguished. In contrast, Russell came to promote a kind of philosophy that was more practically oriented and that needed to be conducted in close connection with the empirical sciences. An early statement of these ideas can be found in *The Problems of Philosophy*, from 1912 (Russell, 1912). Logic, according to Russell, was the most useful tool that philosophy could use.

Russell did his most productive work in philosophy of mathematics around the turn of the century. His early work in this area led to the 1903 *The Principles of Mathematics* (Russell, 1903). He become acquainted with Frege's ideas and his work in logic and the foundations of arithmetic. He discovered a paradox in Frege's *Grundgesetze* (see above), but adopted the logicistic programme developed by Frege and, together with his one-time mentor Alfred North Whitehead, pro-

duced the monumental *Principia Mathematica*, which appeared between 1910 and 1913 (Russell & Whitehead, 1910, 1912, 1913).

After the publication of *Principia* Russell turned to more general philosophical problems in ontology and epistemology, and to politics and social issues. In his philosophical work, he adopted various views, from an rather physicalistic ontology in his early philosophy of logical atomism to the later view of neutral monism, according to which the physical and the mental are on a par, being two different organisations of the same neutral ontological base. His views on how we can acquire knowledge of reality likewise shift. And they continued to do so over long years that Russell was philosophically active. An attribution of particular views to Russell therefore always has to be temporally qualified. But a constant throughout the various stages of Russell's philosophical development is his conviction that the role of logic and of logical analysis is central to all philosophy.

Russell did not just write about political and social issues, he was also actively engaged in these matters. He was a conscious objector in World War 1, stood for parliament but wasn't elected, visited Russia, China and Japan, opened an experimental school, and after World War 2 actively opposed the nuclear policies of both western and eastern nations during the cold war period. He received the Nobel prize for literature in 1950. and over all the years also managed to have a very productive academic career. He published a large number of books and articles, lectured extensively in various countries, notably the US, and held academic positions in Cambridge and in Chicago. His *A History of Western Philosophy* (Russell, 1945), which was published in 1945, became one of the most widely read books on the topic.

The fragments that are treated below stem from Russell's early period, and highlight his rejection of speculative philosophy and his insistence of logic as 'the essence of philosophy'. Another important contribution of Russell's, viz., his analysis of definite descriptions and the distinction between knowledge by acquaintance and knowledge by description will be treated in chapter 17.

2.3 The texts

Frege, Preface from Begriffsschrift

The text comes from the preface of Frege's seminal work, the *Begriffsschrift*. In a few pages Frege provides an insight into his motives for developing the formal system that is presented in the book. These concern mainly the nature of natural language and its role in science and in philosophy, and the contrast with the kind of formal language that Frege develops.

Frege's exposition seems straightforward, but on closer inspection we can distinguish in the text three slightly different perspectives on the relationship between natural language and formal languages, each motivated by different considerations.[10] The three perspectives can be la belled 'extension', 'improvement' and 'reform'.

From the perspective of 'extension' one views a formal language as a means to extend the expressive powe of a natural language. This happens, e.g., when we add elements of symbolic systems, such as calculus or the periodic system.

The perspective of 'improvement' is slightly different: one considers the way in which a natural language expresses certain things to be defective in some sense, and provides a formal language to it better. Precision, or explicitness are main considerations. Think of notation for basic arithmetic: it can be done in natural language, but using the formal notation makes things more easy and surveyable.

The third perspective is that of 'reform'. This is when one considers certain aspects of natural language not just to be lacking in expressive power, or cumbersome, but actually misleading and as such an obstacle in the way of proper philosophical and/or scientific investigation. This perspective is closely related to the conception of an ideal language.

In the preface of *Begriffsschrift* Frege writes about his motivations to move from the use of natural language to the definition and application of a fully formal language at several points. And each of the three perspectives seems to be represented at some point in his exposition.

Thus when Frege writes:

> . . . I found the inadequacy of language to be an obstacle;
> no matter how unwieldy the expressions I was ready to

> accept, I was less and less able, as the relations became more complex, to attain the precision that my purpose required. This deficiency led me to the idea of the present ideography.

he seems to be taking the first perspective: natural language lacks the expressive power to express things with the precision that is needed.

Then there is the comparison of a natural language to the eye and a formal language to the microscope. Here the second perspective, that of 'improvement', seems to be taken. The eye is versatile but lacks sharpness, the microscope provides sharpness but is 'a one-trick pony'.

The third perspective, that of reform, seems the one that inspires the passage where Frege refers to philosophy and its task 'to break the domination of the word over the human spirit'. That domination is due to 'the misconceptions that through the use of language almost unavoidably arise', and it can exposed by using a formal language. Frege was certainly not the first to lament the inadequacy of natural language for proper philosophical analysis and scientific thought. But unlike his predecessors, he had something to back it up: his 'conceptual notation', the first fully explicit example of a formal logical system with sufficient expressive power.

Russell, selections from Our Knowledge of the External World

The selections come from *Our Knowledge of the External World* (Russell, 1914), a compilation of the lectures Russell gave during a visit to the Lowell Institute in Boston. John Slater's introduction gives a detailed overview of the context in which the lectures were written.

In the first chapter we find Russell's criticisms of philosophy as it has been done in the past. Note Russell's use of the term 'philosophy' indicates the western tradition only. Also it is remarkable that he gives a very broad characterisation: there is the classical tradition; evolutionism; and the new kind of philosophy that he himself espouses: philosophy that uses the logical-analytic method.

Russell's main criticism of the classical tradition is connected to what he deems to be the goal of any form of intellectual engagement with the world, i.e., to produce knowledge. That is where the classical tradition fails:

> [...] the great systems of the past serve a very useful purpose, and are abundantly worthy of study. But something different is required if philosophy is to become a science [...]"

Interestingly, Russell does acknowledge that rational reasoning, and thus logic, does play an important role in the classical tradition. So it is not a lack of logic, but a wrong use of it, that is the problem. The difference between the traditional, and wrong use, and the right use resides in how logic interacts with experience, i.e., with empirical knowledge.

'Evolutionism' is the term Russell uses for what he considers the trend of his time: a philosophy based on views that are borrowed from biology, in particular Darwin's theory of evolution. It favours an organic, holistic and continuous, and evolving view of the world, in contrast to the mechanistic and static view that can be associated with classical mechanics.

Russell's criticism is two-fold. He rejects the claim of evolutionism that its world-view is supported by biology; and he objects to what he views is an exclusive focus on moral issues. So, evolutionism is not sufficiently in accordance with experience, i.e., what science tells us about the world, and it does not deal with the kind of questions that Russell thinks philosophy should address, i.e., it does not produce the kind of knowledge that it is the task of philosophy to supply. This is connected with his rejection of the Bergsonian claim that, next to intellect, intuition is a reliable source of knowledge.

The second chapter is devoted to an exposition and defence of the claim that logic is the essence of philosophy, i.e., that:

> [...] every philosophical problem, when it is subjected to the necessary analysis and purification, is found either to be not really philosophical at all, or else to be, in the sense in which we are using the word, logical.

Russell is quick to note that the exact meaning of this claim depends on what is understood by 'logic'. He dismisses classical, Aristotelian logic and the scholastic variants out of hand, and Bacon's inductive method is not regarded as necessary for 'a perfected science'. Russell investigates in some detail the views on induction of John Stuart Mill,

who ties induction to the law of causality. The upshot of Russell's discussion is that logical principles as such can *not* be derived from experience.

Hegel's conception of logic, according to Russell, is a kind of speculative metaphysics that aims to uncover fundamental properties of the world by purely *a priori* reasoning. Although it claims to be a departure from classical, Aristotelian logic, it is not, Russell claims.

Russel then sets out to show that the first real departure from the traditional logic of the western tradition came about with the advent of formal, mathematical logic, as developed by Peano and Frege. Their crucial insight was that not all judgements are of the Aristotelian subject – predicate form, ascribing a property to an entity. Relational judgments and quantified judgments have a different logical structure, and thus warrant different ontological conclusions.

Thus, the key element in Russell's analysis of the philosophical importance of mathematical logic is the notion of logical form. The logical form of a proposition is not a constituent of it, but the way in which its constituents are put together. Logical form in this sense is what philosophical logic is all about. Knowledge of logical forms is the basis of all understanding, and of all inference. The task of philosophical logic is 'to extract this knowledge from its concrete integuments, and to render it explicit and pure.'

Russell illustrates this with an analysis of relational judgments and of various logical properties of relations, emphasising the crucial role they play in scientific and everyday reasoning. He then goes on to discuss how various logical forms determine facts, assertions and propositions. He distinguishes between atomic, molecular, and general logical forms, at each level. Our knowledge of these forms is a form of self-evident, general knowledge that is essential for reasoning, and that cannot be explained as a result of sensory experience, but must rather be considered as a precondition for it.

Thus for Russell logic, the new logic that is, 'has the effect of enlarging our abstract imagination'. The value of its application in philosophy is that 'it [makes] possible at last to see what kinds of problems may be capable of solution, and what kinds must be abandoned as beyond human powers.'

2.4 Study and discussion questions

Study questions

- Which division does Frege make in 'truths that require justification'? And where in that division are the truths of arithmetic?
- What requirements should a language that is suited for expressing logical inference meet according to Frege?
- Wherein does Frege locate the innovation of his formal language?
- In which two directions does Frege see possible extensions of the application of his formal language? And what additions would be needed for those?
- How does Russell evaluate the 'logical-analytic' method in comparison with traditional methods?
- What does Russell mean with the opposition between 'construction' and 'inference' ?
- Which two main sources does Russell regard as the roots of the classical tradition in philosophy?
- What is the gist of Bradley's argumentation against the reality of relations? (see the passage quoted on p. 4–5.)
- What difference does Russell claim there is between the way logic is used in the classical tradition and the role it plays in a scientific philosophy?
- What distinction does Russell make between fact, assertion, and proposition?

Discussion questions

- Can you think of examples of philosophical problems that involve misunderstandings that are related to natural language?
- And how would a formal language (of logic, of mathematics, or a programming language, . . .) be able to help?

- Are all philosophical problems amenable to rational treatment? What are the characteristics of problems that are not?

2.5 Further reading

Michael Dummett's books on Frege, Dummett (1973a, 1985, 1992), though contested in various ways, remain excellent introductions to the various elements of Frege's thought. For an early alternative approach, see Sluga (1980); a good collection of recent work on Frege is Ricketts & Potter (2010).

An early but still relevant introduction to Russell is Pears (1967); see also the more recent Griffin (2003).

On the origins of the analytic tradition and the role of Frege and Russell, see Soames (2014). On the 'pre-'history of modern logic in relation to philosophy, see Lapointe (2018).

Recent work on the relevance of logic to philosophy includes Williamson (2013).